Jia Yi

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PROFESSIONAL SUMMARY

Engineer with over 2.5 years of experience in failure analysis, reliability evaluation, and materials characterization across semiconductor devices, electronic assemblies, and MLCC products. Applies electrical testing, CSAM, LIT, FIB, and SEM/EDS to localize failures, characterize physical defects, and analyze failure mechanisms.

EDUCATION

Sep 2020 – Jun 2023	**Xi'an Jiaotong University**	**M.S. in Chemistry**
Sep 2016 – Jun 2020	**Xiangtan University**	**B.S. in Materials Science and Engineering**

WORK EXPERIENCE

Failure Analysis Engineer (Contractor at Apple)

TEKsystems, Inc. **Aug 2024 – Jan 2026**

- Executed 50+ failure investigations from DVT through mass production for semiconductor devices and electronic assemblies, covering sample preparation, failure localization, cross-sectional analysis, and failure verification.
- Applied I-V testing, CSAM, LIT, FIB, and SEM/EDS to localize and characterize chip cracks, package delamination, solder joint failures, and material-related anomalies.
- Analyzed failure mechanisms by correlating crack morphology and propagation, interface integrity, and localized elemental composition with device architecture and manufacturing process data.
- Developed a CSAM SOP covering sample preparation, scan parameters, and result interpretation; supported SEM and CSAM calibration, equipment verification, and troubleshooting.
- Coordinated overseas engineering teams and local vendors during flash memory test system integration and functional verification to address installation, equipment alarms, and parameter configuration issues.

Complex Module Heat Soak Test Failure Analysis | 2025

- Investigated localized functional abnormalities in a pressure-sensing module after Heat Soak testing and developed a staged localization strategy using CSAM, LIT, cross-section analysis, and targeted FIB verification.
- Correlated evidence across multiple characterization methods, narrowing the suspected failure region by approximately 90% and defining the direction for subsequent failure-mechanism verification and improvement studies.

Process Engineer

Yuanliu Hongyuan (Suzhou) Electronics Technology Co., Ltd. **Jul 2023 – Jun 2024**

- Developed and validated 10+ MLCC products through design, trial production, and qualification, evaluating electrical performance, reliability, and manufacturing stability; three products advanced to mass production.
- Evaluated 10+ MLCC batches through TC and HALT testing, then used DPA and SEM to characterize electrode delamination, ceramic microcracks, and related failure modes in abnormal samples.
- Used Minitab to correlate process parameters with capacitance distribution, identified ceramic powder type as a key contributor to capacitance variation, and informed process adjustments that increased trial-production yield by 3%.

TECHNICAL SKILLS

Failure Analysis: OM, CSAM, SEM/EDS, LIT, FIB, DPA

Reliability Testing: Thermal Cycling (TC), HALT

Languages: English (Technical Communication, Report Writing), Chinese (Native)